

Table 1 Effect of (wildfire and non-wildfire) PM _{2.5} on respiratory hospital admissions.									
Fire upwind + strong SAW (1999–2012)		Regression model for respiratory admissions (rate per 100,000 people)							
Aggregated sources (smoke and non-smoke)		Approach used to isolate wildfire-specific PM _{2.5}							
		Instrumental Variable		Imputation		Interaction		Seasonal Interpolation	
		Wildfire-specific	Non-smoke	Wildfire-specific	Non-smoke	Wildfire-specific	Non-smoke	Wildfire-specific	Non-smoke
PM _{2.5} coefficient (95% CI)		0.0014 (0.00077–0.0021)	0.0013 (0.00068–0.0020)	0.018 (0.0064–0.030)	1.00068 (1.00049–1.00087)	1.00061 (0.10–1.0015)	0.0024 (0.0018–0.0030)	0.0055 (–0.00068 to 0.012)	
% change with 10 µg m ^{–3} PM _{2.5} (95% CI)		3.8 (–1.2 to 8.9)	0.72 (0.36–1.1)	10 (3.5–16.5)	0.67 (0.48–0.86)	1.28 (0.37–2.19)	1.3 (0.97–1.7)	3.0 (–0.37 to 6.3)	
All regressions include controls:flu admissions, weather covariates, day-of-week effects, month-of-year effects, zip code fixed effects, and a time trend. Summer months (June, July, August) are excluded. Mean PM _{2.5} = 15.6 µg m ^{–3} (IQR = 9.2 µg m ^{–3}). Mean rate of respiratory admissions per 100,000 people = 1.85.									

times greater than equivalent doses of ambient PM¹⁴. A subsequent study by Wegesser et al.⁴⁴, expanded on these findings to show that substances such as polycyclic aromatic hydrocarbons can be present in much higher concentrations in smoke versus levels detected in ambient air. Another study⁴⁵ examined the inflammatory responses due to wildfire smoke PM exposure and found significant changes in reactive oxygen species and subsequent oxidative stress, leading to higher cell degeneration and potential programmed cell death.

In addition to differences in the chemical composition of smoke and ambient PM, different stages of biomass combustion appear to have differential impacts on health¹⁵. Recent findings have suggested that the types of trees and the temperature at which the combustion takes place may explain the differential toxicity regarding wildfire-specific PM, as observed in mouse lung response¹⁵. All the above evidence suggests that the assumption that all particles of a given size class have the same toxicity (which is currently the basis for regulation of airborne PM_{2.5}) may be inaccurate. Future studies should address the epidemiological response to wildfires affecting different ecosystems and fuel types and burning at different combustion temperatures.

Understanding the impacts of wildfire on public health is of vital importance in Southern California where several factors may increase exposure to wildfire-specific PM_{2.5} in the context of global climate change. Wildfire severity and risk in this region will likely intensify in the warming future⁴⁶ as changing precipitation and wind patterns gradually push the wildfire season from fall to winter when back-to-back SAWs can cause wildfires to burn longer^{29,47,48}. In addition, given that most large fires in Southern California are caused by human ignitions, whether accidental or deliberate, the current and projected population growth trends and the expansion of the Wildland-Urban Interface may create additional wildfire ignitions⁴⁹ in the region. Our results could be transferred to similar regions in the US and the world where wind-driven wildfires cause damage to public health (via smoke PM_{2.5}) and property, particularly in a changing world scenario where wildfire-PM_{2.5} is projected to increase relative to emissions from other sources.

Methods

The following sections describe the data and methodology used to estimate wildfire-specific PM_{2.5} and to quantify its impact on respiratory health. We used ArcGIS 10.5⁵⁰, R version 3.5.1⁵¹, and Stata version 16⁵² for all analyses detailed below.

Respiratory health. Daily hospital admissions for respiratory diseases were obtained from the California Office of Statewide Health Planning and Development (OSHPD) database of patient discharge data for the study period and spatial domain. Respiratory hospitalizations correspond to the ICD 9 codes 460:519 which include pulmonary diagnoses, such as asthma, COPD, pneumonia, and interstitial lung disease. In addition, data for flu diagnosis were also available. All data were aggregated at the daily level by zip code and converted to rates of admission by dividing the admission counts by the population.

Fine particulate matter (PM_{2.5}). Daily-, zip code-specific concentrations of PM_{2.5} were estimated from 1999 through 2012³¹ using 24-h daily means sampled and analyzed by the US EPA Air Quality System (<https://www.epa.gov/aqs>) at ground monitoring stations within a 20 km radius of each population-weighted zip code centroid. Values were interpolated using an inverse distance weighting approach³¹, which gives greater importance to monitoring stations closer to the point of interest. These PM_{2.5} values, coming from monitoring station data, represent fine particulate matter from all sources, including ambient levels and wildfire smoke. PM_{2.5} values were available on 578 zip codes in our study region, subject to monitoring station data availability, with varying degrees of missing data. Overall, the mean percentage of missing values in these zip codes was 33% (median: 15%).

Wildfire upwind and strong Santa ana winds. Wildfire exposure was derived using fire perimeters in Southern California from 1999 to 2012 from the Fire and Resource Assessment Program of the California Department of Forestry and Fire Protection (CalFire; <http://frap.fire.ca.gov/>). Zipcodes were assigned a wildfire event upwind if their centroids were within a 160-km radius of the fire perimeter's centroid (Fig. 4) and were located within a range of angles (roughly from –10° to

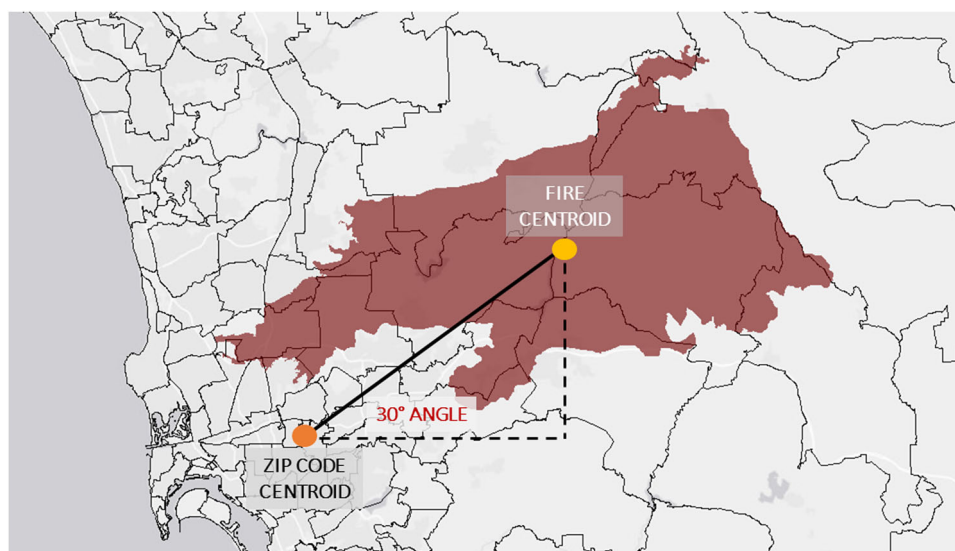


Fig. 4 Wildfire upwind exposure estimation. The relationship between a given zip code centroid and a fire centroid is assessed by means of geospatial tools. Fires burning upwind are considered within the context of the north-easterly direction of Santa Ana Winds. We use the distance between centroids, as well as the angle of the spatial relationship as shown here, to classify a zipcode day as exposed (or not).

120°) that capture the north-easterly direction and the smoke plumes associated with Santa Ana winds. Neighboring zip codes to the fire perimeter were also assigned a wildfire event. Wildfire conditions were defined as the days during which a wildfire was detected upwind of a given zip code.

The combination of strong SAWs and wildfire upwind is the condition under which smoke is expected to impact communities downwind of a wildfire. We used the daily version of the original hourly SAWRI²⁶, evaluated over the Santa Ana wind domain, to identify days with strong and widespread SAWs. SAWRI, expressed in m s^{-1} , provides an observationally validated regional daily summary of the dynamically downscaled SAWs. Strong Santa Ana wind conditions were defined as any Santa Ana wind day when SAWRI was above 3.06 m s^{-1} (its median value, conditional on taking a positive value).

Smoke plumes. Smoke plumes were obtained from the NOAA Hazard Mapping System (HMS), available from September 2005 onward. The HMS product uses visible satellite imagery and trained satellite analyst skills to estimate the spatial extent of smoke, though it cannot discern whether a given plume is at ground level or higher in the atmosphere⁵³. In addition, the HMS smoke-plume extent data has not been validated and could thus have systematic biases because discrimination of smoke can vary by region, season, and weather conditions⁵⁴. However, HMS smoke plumes remain a common binary metric used to determine if smoke is present in the atmospheric column on a given day³⁵. The HMS smoke products are stored as polygon shapefiles representing the spatial extent of daily smoke plumes (http://satpessanone.nesdis.noaa.gov/volcano/FIRE/HMS_ARCHIVE/). A simple smoke binary variable was created by intersecting zip code polygons with smoke polygons, which was then used as an indication of daily exposure to wildfire $\text{PM}_{2.5}$. We included the additional condition of wildfire presence within a 160-km radius from a given zip code in order to classify it as exposed to wildfire smoke.

Weather covariates. We collected hourly data from NOAA's National Centers for Environmental Information Integrated Surface Database (NCEI ISD; <https://www.ncdc.noaa.gov/isd/>) and calculated 24-h daily means for wind speed, temperature, and humidity. Values were interpolated using an inverse distance weighting approach³¹ and considering the daily observations from monitoring stations within a 20 km radius of each population-weighted zip code centroid.

Estimating wildfire-specific $\text{PM}_{2.5}$

Instrumental variable (IV) approach: a two-stage regression. We use a novel joint instrument within a two-stage regression based on both wind and wildfire occurrence. Specifically, we modeled the incidence of wildfire upwind during strong Santa Ana winds, in order to isolate the effects of wildfire-specific $\text{PM}_{2.5}$ on respiratory admissions. Indeed, through using these joint instruments and a two-stage least square approach, we estimate the local average treatment effect of $\text{PM}_{2.5}$ on respiratory hospital admissions⁵⁵. Said differently, this procedure allows us to isolate “complier $\text{PM}_{2.5}$ values” regarding the presence an upwind wildfire smoke in a given zip code-day. In this context, monotonicity implies that the joint instruments have no effect on $\text{PM}_{2.5}$ levels on non-wildfire zip code-days and that all zip

code-days with upwind wildfire are affected in the same way. Furthermore, using this approach we can assume that any effect the joint IV has on the hospital admissions is only mediated by a “local” variation of $\text{PM}_{2.5}$ for eligible zip code-days only (exclusion restriction criteria) while having a strong correlation between the joint IV and $\text{PM}_{2.5}$ levels.

In the first stage (Eq. 1, below), $\text{PM}_{2.5}$ is regressed on a binary variable combining the presence of wildfire upwind and strong SAWs. In the second stage (Eq. 2), rates of respiratory admissions per 100,000 individuals are regressed on the fitted values of the explanatory variables, including wildfire $\text{PM}_{2.5}$ estimated in the previous stage. For the IV estimation to be consistent, all exogenous variables used in the second stage must also be included in the first stage and only one exogenous variable coefficient may be estimated in the second stage for each instrumental variable included in the first stage regression⁵⁶. The following controls were included: the daily number of flu admissions by zip code, weather covariates: mean daily wind speed, temperature and humidity, day-of-week effects, month-of-year effects, a linear time trend, and zip code fixed effects. Ordinary least squares regressions were implemented with the plm R-package for panel data⁵⁷.

First stage regression

$$\text{PM}_{2.5it} = \gamma_0 + \gamma_1 \text{Exposure Definition}_{it} + \gamma_2 \text{Flu}_{it} + \gamma_3 \text{Wind}_{it} + \gamma_4 \text{Temperature}_{it} + \gamma_5 \text{Humidity}_{it} + \text{Zip}_i + \text{Weekday}_t + \text{Month}_t + \tau t + u_{it} \quad (1)$$

Second stage regression

$$\text{Resp}_{it} = \beta_0 + \beta_1 \widehat{\text{PM}}_{2.5it} + \beta_2 \text{Flu}_{it} + \beta_3 \text{Wind}_{it} + \beta_4 \text{Temperature}_{it} + \beta_5 \text{Humidity}_{it} + \text{Zip}_i + \text{Weekday}_t + \text{Month}_t + \tau t + \epsilon_{it} \quad (2)$$

Spatio-temporal multiple imputation approach. For this approach, we used a cubic spline interpolation to impute the $\text{PM}_{2.5}$ concentrations attributable to non-smoke sources in zip code/days identified as exposed to wildfire smoke. Cubic splines are an extension of polynomial regression where times t are divided into k intervals called knots. For each interval, a regression is fit with three parameters. This method has been found to allow the inclusion of local characteristics of a trend without prejudicing its global characteristics.

More specifically, we followed the steps below:

- Using the exposure definition of wildfire upwind and strong SAWs (or smoke plumes; Sections 3 and 4), we identified the zip code days exposed to wildfire smoke in our original $\text{PM}_{2.5}$ dataset.
- We used a spline interpolation approach to impute the values of non-smoke $\text{PM}_{2.5}$ on all zip code days categorized as exposed to smoke and where $\text{PM}_{2.5}$ data were originally available (i.e., we did not impute missing values in the original dataset). Cubic spline interpolation was implemented in R by means of the imputeTS package⁵⁸. This step provided estimates of ambient $\text{PM}_{2.5}$ unrelated to wildfire smoke.
- We then subtracted all non-smoke $\text{PM}_{2.5}$ values from the original daily $\text{PM}_{2.5}$ concentrations to obtain the levels of $\text{PM}_{2.5}$ attributable to wildfire smoke in zip code days previously categorized as exposed.

- d. Lastly, rates of respiratory admissions are regressed on the wildfire-specific $PM_{2.5}$ concentrations (Eq. 3), including controls for the daily number of flu admissions by zip code, day-of-week effects, month-of-year effects, a linear time trend, and zip code fixed effects.

$$Resp_{it} = \beta_0 + \beta_1 Wildfire_{it} - \beta_2 PM_{2.5, it} + \beta_3 Flu_{it} + \beta_4 Wind_{it} + \beta_5 Temperature_{it} + \beta_6 Humidity_{it} + \beta_7 Zip_i + \beta_8 Weekday_t + \beta_9 Month_t + \tau t + \epsilon_{it} \quad (3)$$

Interaction model. We used a zip code fixed effects Poisson regression model to quantify the effects of wildfire-specific and non-wildfire-specific $PM_{2.5}$ on respiratory admissions. The rate of admissions was modeled as a function of the interaction between wildfire exposure and $PM_{2.5}$ and a set of control variables comprised of the number of flu admissions, mean daily wind speed, mean daily temperature, mean daily humidity, zip code fixed effects, dummy variables for day of week and month of the year, a linear time trend, and log-transformed zip code-specific population as an offset term (Eq. 4).

$$Resp_{it} = \exp(\log(Population_{it})) + \beta_1 PM_{2.5, it} + \beta_2 Wildfire_{it} + \beta_3 PM_{2.5, it} \times Wildfire_{it} + \beta_4 Flu_{it} + \beta_5 Wind_{it} + \beta_6 Temperature_{it} + \beta_7 Humidity_{it} + \beta_8 Zip_i + \beta_9 Weekday_t + \beta_{10} Month_t + \tau t + \epsilon_{it} \quad (4)$$

Using this framework, the effect of non-wildfire $PM_{2.5}$ is captured by β_1 , and the effect of wildfire $PM_{2.5}$ is captured by the sum of β_1 and β_3 , which are then transformed to obtain the marginal effects of changes in non-wildfire-specific and wildfire-specific $PM_{2.5}$. Standard errors and confidence intervals on the marginal effects are obtained using the delta method.

Seasonal interpolation. We based this approach on the method implemented by Lipner et al.²⁷ to segregate wildfire smoke $PM_{2.5}$ from other sources of $PM_{2.5}$:

- As in the imputation approach, we identified the zip code days exposed to wildfire smoke in our original $PM_{2.5}$ dataset.
- Subsequently, daily non-smoke $PM_{2.5}$ was provisionally estimated by means of inverse distance weighting spatial interpolation considering $PM_{2.5}$ data from only zip code days not exposed to wildfire smoke.
- Seasonal non-smoke $PM_{2.5}$ was then calculated as the median of the non-smoke $PM_{2.5}$ estimates above, for each three-month season and grid cell.
- These seasonal non-smoke $PM_{2.5}$ concentrations were subtracted from the original $PM_{2.5}$ dataset (setting negative differences to zero) to compute $PM_{2.5}$ derived from wildfire smoke. The remaining component of daily $PM_{2.5}$ (equal to the seasonal background unless the background was higher than the full daily $PM_{2.5}$ concentration) was then defined to be the zip code-specific, daily non-smoke $PM_{2.5}$ concentration.
- Rates of respiratory admissions are regressed on the wildfire-specific $PM_{2.5}$ concentrations (Eq. 5), including controls for the daily number of flu admissions by zip code, day-of-week effects, month-of-year effects, a linear time trend, and zip code fixed effects.

$$Resp_{it} = \beta_0 + \beta_1 Wildfire_{it} - \beta_2 PM_{2.5, it} + \beta_3 Flu_{it} + \beta_4 Zip_i + \beta_5 Weekday_t + \beta_6 Month_t + \tau t + \epsilon_{it} \quad (5)$$

Reporting summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability

The data, with the exception of health data, that support the findings of this study are available from the corresponding author upon reasonable request. Publicly available data is also found here: daily mean $PM_{2.5}$ concentrations at EPA monitoring sites: <https://www.epa.gov/aqs>; CalFire Wildfire Perimeters: <http://frap.fire.ca.gov>; NOAA Hazard Mapping System smoke plumes: ftp://satsepnone.nesdis.noaa.gov/volcano/FIRE/HMS_ARCHIVE/; meteorological variables from NOAA's National Centers for Environmental Information Integrated Surface Database: <https://www.ncdc.noaa.gov/isd>.

Code availability

Scripts that support the findings of this study are available from the corresponding author upon reasonable request.

Received: 24 February 2020; Accepted: 3 February 2021;

Published online: 05 March 2021

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Acknowledgements

This work was funded by the University of California Office of the President via Multicampus Research Programs and Initiatives (MRPI; MRP-17-446315) and the National Oceanic and Atmospheric Administration's Regional Integrated Sciences and Assessments (RISA) California–Nevada Climate Applications Program award NA17OAR4310284. Additional support was provided in part by the Alzheimer's Disease Resource Center for advancing Minority Aging Research at the University of California San Diego (P30AG059299 National Institute on Aging) and R01CA228147-01A1 (National Cancer Institute). This work was also supported by Office of Environmental Health Hazard Assessment #19-E0022.

Author contributions

R.A. and T.C. contributed equally to this paper: both authors carried out the data analysis and R.A. led the writing and the revision of the paper. A.G. and T.B. conceived the project and assisted in writing the paper. All authors designed the study, discussed the results, and commented on the final paper.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-021-21708-0>.

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Peer review information *Nature Communications* thanks Xerxes Seposo and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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